Fiche cours N1

ANGLAIS ELECTROMÉCANIQUE

PROPERTIES OF MATERIALS

Objectif:

By the end of this lesson, students will be able to:

- Identify and define key material properties.
- Explain how different materials are suited for specific purposes based on their properties.
- Test and compare material properties through simple investigations.

Ce que tu vas apprendre:

Materials are the substances that objects are made from. Different materials have different properties, which means they behave differently depending on what they are used for. Understanding these properties helps us choose the right material for a specific job.

Material Properties their Definitions and Uses:

PROPERTIES	DEFINITIONS	USES
RESILIENCE	The ability of a material to absorb energy when it is elastically deformed and release that energy upon unloading, without permanent deformation.	Springs, vibration dampers, protective visors, sports equipment.
DUCTILITY	The ability of a material to stretch or be drawn out into a thin wire without breaking when subjected to tensile (pulling) stress.	Electrical wiring, metal cables, deep-drawn parts.
TENACITY (TOUGHNESS)	The ability of a material to absorb energy and plastically deform without fracturing; resistance to impact and crack propagation.	Helmets, body armor, car frames, safety equipment.
ELASTICITY	The ability of a material to return to its original shape and size after the stress causing deformation is removed.	Springs, seals, shock absorbers, elastic bands
PLASTICITY	The ability of a material to undergo permanent deformation without breaking when a load exceeds its elastic limit.	Metal stamping, molding plastic parts, can forming.
MALLEABILITY	The ability of a material to withstand compressive forces and be deformed (flattened or hammered) into thin sheets without cracking.	Metal sheets, foils, coin production, architectural cladding.
HARDNESS	The ability of a material to resist surface indentation, scratching, or abrasion.	Cutting tools (drills, saw blades) • Wear-resistant coatings • Ball bearings • Armor plating
CONDUCTIVITY	The ability of a material to allow the flow of energy such as **electricity** or **heat** through it.	**Electrical wiring** • Electronic circuits • Heat exchangers • Solar panels.

Exemples:

EXEMPLE 1: Ductile Materials
- Copper
- Aluminum
- Gold

EXEMPLE 2: Resilient Materials
- Rubber
- Titanium
- Polycarbonate

EXEMPLE 3: Malleable Materials
- Gold
- Silver
- Copper

Astuces pour progresser ou conseils

Learning English is like building a bridge to the world. Every new word you learn, every sentence you practice, adds another stone to that bridge.
At first, it may feel slow and difficult, but step by step, you are building a powerful path that will open doors to better jobs, travel, friendships, and new ideas.
Don't be afraid to make mistakes, mistakes are proof that you are trying.
Every great speaker was once a beginner. Stay curious, stay brave, and remember: the more you practice, the stronger and more confident you will become.

Practice makes perfect